
Supplementary information

Observation of universal ageing dynamics in antibiotic persistence

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Supplementary Experimental Materials and Methods

Antibiotic survival assay following a stress exposure. Bacteria were grown in a 96-well plate and subjected to a stress condition, as described in Methods – *Culture conditions and chemicals*. At several time points, samples of the culture were diluted into liquid M9 medium containing ampicillin (100 µg/mL; Sigma) +1% Casamino acids +kanamycin (23 µg/mL), and incubated at 32°C. Dilution from SHX and sodium azide was 1:100 and 1:500, respectively. The ampicillin exposure duration was 9.5 hours. After the exposure, penicillinase (Sigma) was added to a final concentration of 1 unit/mL. Colony- forming unit (CFU) assays were performed both before and after the exposure using LB-agar plates supplemented with kanamycin (23 µg/mL). The viability of the plated culture was estimated based on colony counts (CFU) by the maximum likelihood estimator⁵³.

Strains and plasmids construction. *E. coli* strain DH5αZ1 was obtained from H. Bujard⁵⁴. A P1_{vir} lysate of DH5αZ1 was used to transduce KLY⁷ into KLYR, under a spectinomycin resistance screen, to move the *lacI* and *tetR* genes into the KLY background under constitutive promoters. The *relA1*⁵⁵ and *spoT1*⁵⁶ mutations of KLYR confer a relaxed phenotype^{56,57}.

Table S1: Strains and plasmids

Strain/plasmid	Relevant characteristics	Source	Doubling time ^a
KLY	<i>relA1 spoT1</i> relaxed phenotype ^{56,57}	⁷	
KLYR/pZA21mCherry	KLYR: A P1 _{vir} lysate of DH5αZ1 ⁵⁴ was used to transduce KLY ⁷ into KLYR, under a spectinomycin resistance screen, to move the <i>lacI</i> and <i>tetR</i> genes into the KLY background under constitutive promoters.	This work	50 ± 3 min (M9) 28 ± 1 min (LB) ^b
MG1655/pZA21RmCherry	MG1655 (CGSC #6300); pZA21RmCherry constructed by cloning <i>tetR</i> after <i>kanR</i> (transcriptional fusion) in pZA21mCherry plasmid	This work	62 ± 2 min (M9)
KLY <i>metG</i> ^T	Evolution E1 – tolerance-by-lag	⁵⁸	
EPEC (E2348/69)	EPEC wild type (strain O127:H6)	⁵⁹	24.7 ± 0.2 min (LB)

MGY $\Delta motA$	MGY ⁶⁰ $\Delta motA$ was constructed by λ Red ⁽⁶¹⁾ , by insertion of amplified <i>motA::kan^R</i> fragment from Keio collection ⁶⁹ strain #JW1879 into MGY	This work	
KLY $\Delta motA$	KLY $\Delta motA$ was constructed by λ Red ⁽⁶¹⁾ , by insertion of amplified <i>motA::kan^R</i> fragment from Keio collection ⁶⁹ strain #JW1879 into KLY	This work	
pZA21mCherry plasmid	Plasmid encoding fluorescent protein mCherry under <i>tet</i> promoter (PLtetO1)	⁶²	

a From OD measurements in liquid M9 medium or LB at 32°C (see Methods – *Culture conditions and chemicals*). The standard deviation is over 3 biological replicates (each averaged over at least 4 technical replicates), except where otherwise indicated.

b From OD measurements in liquid LB at 32°C (see Methods – *Culture conditions and chemicals*). The standard deviation is over 10 technical replicates.

Definition of terms in analysis of single-cells fate (Fig. 2e).

1. Arrested: Cells that do not divide or elongate (most cells freeze within 1 hour after SHX addition)
2. Divide: Cells that divide once after 2 hours of SHX exposure. No cell divided more than once under SHX.
3. Lyse: Cells that lyse, losing their YFP signal.
4. Resume: Cells that divide after SHX is washed with fresh medium.

During recovery, division of some cells caused about half of the cells to be pushed out of the channels before being counted.

Viability significance tests. The viability (CFU) was measured at various time points. The data from separate experiments were collected and linearly interpolated to equally spaced times. Two-sided Student's paired *t*-tests were performed between $t = 0$ and later times.

Goodness of fit tests for relaxation dynamics. From each experiment, one lag time distribution was selected. The selected curve was after the saturation of ageing. The curves were linearly interpolated at regular times to enable averaging the 1 – cdf over the replicates, and calculating standard deviation. Then, the average was fitted to three different functional forms, with the inverse variances as weights to the fit. The functional forms were:

Exponential: $f(t) = C \exp(-\beta t)$

Stretched exponential: $f(t) = \exp(-Ct^\beta)$

Powerlaw: $f(t) = Ct^\alpha$

For the same condition, the range fitted was the same for all functional forms. The stretched exponential function intersects ($t = 0$; $f(0) = 1$). Therefore, the fit was performed after shifting the curves, if needed.

The goodness of fit is given by the reduced χ^2

$$\chi_{\text{red}}^2 = \frac{1}{N-2} \sum_{i=1}^N \frac{(y_i - f(t_i))^2}{\sigma_i^2}$$

Where (t_i, y_i) and σ_i are the experimental data and standard deviation, and N is the number of data points. The results are summarized in Table S3.

Statistics and Reproducibility

All experiments in main figures were reproduced at least three times (biological replicates) with similar results. Only the following experiments include less than three biological replicates:

1. The microfluidic experiment shown in Extended Data Fig. 6 was performed once.
2. The antibiotic survival assay shown in Extended Data Fig. 7d include two biological replicates with similar results.
3. The saline starvation experiment in Extended Data Fig. 9 includes two biological replicates.

Theory: Randomly Connected Cycles Network Model (RCCN)

Our goal was to reproduce the ageing dynamics observed experimentally in the recovery of single bacteria exposed to acute stress that arrests growth, such as serine hydroxamate (SHX) or sodium azide. The genetic and metabolic networks required for the regrowth of the bacteria, after these stresses are removed, involve a large part of the intracellular network. This network of interacting cellular compounds (proteins, metabolites, RNAs, etc.) is modelled below by analogy with models displaying physical ageing but is used as a toy model, without aiming at reproducing specific cellular pathways

Ageing dynamics in physical systems is one of the hallmarks of glasses and spin glasses. The simplest model exhibiting this feature is the Sherrington and Kirkpatrick model²³ which consists of large number of spins, s_i , $i = 1, 2 \dots N$ (that can be regarded as Boolean variables, $s_i = \pm 1$) with random and symmetric coupling $J_{ij} = J_{ji}$. The discrete time zero temperature Glauber dynamics of this system is given by

$$s_i(t+1) = \text{sign} \left(\sum_{j \neq i} J_{ij} s_j(t) + H(t) + \xi(t) \right), \quad (\text{S1})$$

where $H(t)$ is an external magnetic field, and $\xi(t)$ is short ranged correlated external noise. Ageing in this system is observed, for instance, when the magnetic field is applied for a finite waiting period, T_w , after which the average magnetization $\overline{M}(t) = (1/N) \langle \sum_i s_i(t) \rangle$, averaged over realizations is recorded. In this case, the decay of the magnetization displays a memory of T_w , having a qualitatively different behavior depending on the value of T_w , see [63] and references therein.

At first sight, the time evolution of Boolean variables as given by Eq. (S1) may provide also a crude statistical description for the dynamics of the internal compounds in a living cell, where each Boolean variable represents one chemical compound assumed to be in only one of two possible states – active or inactive (see Extended Data Fig. 2). Boolean description of genetic networks is based on the often observed sigmoidal dependence of gene expression on conditions and has been used to study statistical properties of cellular networks⁶⁴. However, unlike the Sherrington and Kirkpatrick model in which the interaction between two nodes is symmetric, the coupling constants which drive the production of cellular components are asymmetric, i.e. $J_{ij} \neq J_{ji}$. In the asymmetric case, the magnetization decays instantaneously and no ageing is observed (see Extended Data Fig. 4a, and e.g., [65, 66]). Similarly, randomly connected architectures such as the Erdős-Renyi networks do not display ageing (Extended Data Fig. 4b,c). To obtain ageing in the time evolution (Eq. (S1)) with asymmetric coupling constants, some elements of the cell network architecture should be retained. A natural option is choosing these coupling constants to form cycles that are weakly coupled (Fig. 3a), reproducing the inherent recurrence²⁵ of the core networks involved in growth. For example, charged tRNAs are

needed for the production of the enzymes responsible for their charging. Similar recurrences create cycles that may include the several layers of the cellular network i.e. metabolism, transcription, translation etc., and may not be mapped as a cycle in each layer separately.

In our model, termed the “randomly connected cycles network (RCCN)”, the cycles lengths are independent random variables taken from a power law distribution in a finite range between $L_{\min}$ and $L_{\max}$.

$$P(L) = \nu L^{-\alpha}, \quad L_{\min} \leq L \leq L_{\max} \quad (S2)$$

The prefactor, ν , is the normalization constant. All cycles are coupled to each other via one of their sites with asymmetric coupling constants $J_{ij} \neq J_{ji}$, where now the indices i and j refer to cycles. These coupling constants are independent random variables drawn from a Gaussian distribution with zero mean and variance which scales with the number of cycles, N_c , as

$$\langle J_{ij}^2 \rangle = \frac{\gamma^2}{N_c}, \quad (S3)$$

with γ a free parameter. The dynamics of the nodes within each cycle is simple shift dynamics of the node states,

$$s_i^{(k+1)}(t+1) = s_i^{(k)}(t), \quad L_i - 1 \geq k \geq 1 \quad (S4)$$

where $s_i^{(k)}(t)$ denotes spin k of cycle i , of length L_i .

The Glauber-like dynamics of the spins at the connecting nodes of the cycles, which we denote by $s_i^{(0)}$, is given by

$$s_i^{(0)}(t+1) = \text{sign} \left(s_i^{(L_i-1)}(t) + \sum_{j \neq i}^{N_c} J_{ij} s_j^{(0)}(t) + H(t) \right). \quad (S5)$$

The choice of J_{ij} , as independent random variables, needs justification. It is clear that evolution over billions of years has led to some very specific values for the coupling constants, which are designed to optimize the cell's performance under changes in the environmental conditions. However, our assumption is that for acute stresses that drive the cell away from its adaptive potential, the complexity of the system's dynamics can be approximated by random coupling constants.

In the experimental system, isogenic bacteria share the same topology of the network. Therefore, the cycles were fixed for all cells, as well as the sign of the connections between

nodes. The initial conditions affected the strengths of the J_{ij} (not their sign) and reflect the fact that single cells may be in very different states that can affect the way the idealized pair-wise interactions proceed. We note that the results are similar if we lift these biological constraints, and average over arbitrary J_{ij} and network topologies (see Extended Data Fig. 5c). Finally, when $\gamma > 1.3$, the weak external noise has a negligible effect and therefore is set to zero.

Acute stress on the living cell, such as exposure to SHX, which creates an artificial starvation for serine, or to sodium azide, which depletes ATP, is realized in the model by the application of an external magnetic field, which aligns the spins in its direction. This alignment of the spins in the up direction (“OFF”) is analogous to the pleiotropic decrease in the activity of many cellular compounds upon starvation or ATP depletion. Therefore, the macroscopic state of the cell is characterized by the number of active nodes, which is equivalent to the magnetization in the spin model.

The magnetic field induces a net magnetization which we define by

$$M(t) = \frac{1}{N_c} \sum_{i \in \text{cycles}} \frac{1}{L_i} \sum_{k \in i^{\text{th}} \text{cycle}} s_i^{(k)}(t) \quad (\text{S6}).$$

A typical example for the behavior of the magnetization, after the magnetic field is switched off is depicted in the inset of Extended Data Fig. 3b. It exhibits strong fluctuations on top of a slow decay, eventually reaching the initial level before the turning on of the magnetic field. Similarly, cells recovering from the stress period will be able to start growing as soon as the total number of active nodes returns to its original value. Therefore, the lag time for regrowth of each bacterium is similar to the first crossing of the magnetization back to zero, its original value.

Numerical solution

In the experiment, cells are subjected to abrupt stress of duration T_w , after which the time for the recovery in each one of the cells (i.e. the lag time) is recorded. The experimental cumulative distribution function (CDF) of these random time events is evaluated and 1–cdf graphs for various waiting times are plotted, as shown in Fig. 3d.

The analogous procedure for the numerical simulation is as follows. First the cycles lengths are chosen from the distribution (S2) with $\alpha = 1.5$, $L_{\min} = 1$, and $L_{\max} = 2500$, such that the total number of spins in the system is $N = 2^{14}$ (corresponding to $N_c \sim 410$ cycles). Then the coupling constants J_{ij} are set as random independent variables of normal distribution with zero mean and variance given by (S3). One such realization of the network corresponds to a single ‘cell’. Next Eqs. (S4) and (S5) are iterated for 2000 time steps starting from random initial conditions of the spins. The magnetic field, $H = 0.8$, is turned on for a time period T_w after which the system is further iterated for an additional 4000 time steps (Extended Data Fig. 3a). The magnetization (S6) of the system is calculated at each time step. The

distribution function of the first crossing of the magnetization back to zero, is obtained from the ensemble over coupling constants, J_{ij} , and initial conditions. The statistics is gathered from different simulations in a similar amount to the number of single cells measured in the experiment, and the 1-cdf graphs obtained for different T_w is presented in Fig. 3b, showing a qualitative agreement with the experimental plot.

Code availability: All Matlab scripts for the simulation⁵² are available through GitHub at the link: <https://github.com/NQBLab/RCCN>.

Analytical Results

In order to gain intuitive understanding of the RCCN dynamics, a simpler phenomenological description of our model can be derived from a mean field approach⁶⁷. In this description the system comprises of independent cycles where each cycle behaves as a capacitor in series with a resistor which determines its decay time. The latter is proportional to the cycle length. The external magnetic field is analogous to a voltage source that charges the capacitor when it is turned on, and the magnetization corresponds to the charge on the capacitor plates. The capacitors start discharging when the magnetic field is switched off. The dynamics of the average magnetization, $\bar{M}(t) = \langle M(t) \rangle$, where $M(t)$ is defined in Eq.(S6), can be expressed in terms of the exponential integral function, $E_n(x)$,

$$\bar{M}(t) = \sqrt{\frac{2}{\pi}} \frac{H}{\gamma} \left\{ \frac{\nu}{L_{\max}^{\alpha-1}} \left[E_{2-\alpha} \left(\frac{t}{\tau_{\max}} \right) - E_{2-\alpha} \left(\frac{t+T_w}{\tau_{\max}} \right) \right] - \frac{\nu}{L_{\min}^{\alpha-1}} \left[E_{2-\alpha} \left(\frac{t}{\tau_{\min}} \right) - E_{2-\alpha} \left(\frac{t+T_w}{\tau_{\min}} \right) \right] \right\} \quad (\text{S7})$$

where $\tau_{\max} = \tau_1 L_{\max}$, $\tau_{\min} = \tau_1 L_{\min}$ and τ_1 is the time scale of decay of the magnetization of a unit length cycle. In Extended Data Fig. 3b we present the results of the average magnetization of the numerical simulation together with the analytical solution (Eq. S7). The only fitting parameter needed to obtain the analytical results in Extended Data Fig. 3b (black dashed lines) is the decay time $\tau_1 = 1.9$ min.

The approximate solution for the distribution of lag times in this phenomenological model is

$$1 - \text{cdf} = 1 - \beta(t) - \left(\frac{1}{\beta(t)} - 1 \right) \int_0^t dt' \beta(t-t') \frac{1}{\tau} \exp \left(-\frac{t'}{\tau} \right) \quad (\text{S8})$$

where

$$\beta(t) = \frac{1}{2} \operatorname{erfc} \left(\frac{\bar{M}(t)}{\sqrt{2}\sigma} \right). \quad (\text{S9})$$

Here $\text{erfc}(x)$ is the complementary error function and σ , the standard deviation of the magnetization.

In Eq. (S8), τ is the typical time needed for the magnetization, $M(t)$, to return to zero, for the first time, after it crossed zero value from below (see inset of Extended Data Fig. 3b). The analytical solution in Fig. 3b (dashed lines) is obtained by fitting τ appropriately.

The intermediate asymptotic of the survival probability, $1-\text{cdf}$, in the regime $\tau_{\min} \ll t \ll \tau_{\max}$ can be evaluated by taking into account that $\tau \ll t$. In this case the upper limit of the integral (S9) can be extended to infinity while the function $\beta(t-t')$ is expanded in powers of t' . The zeroth order power cancels the first term in (S8), and the first order term yields:

$$1-\text{cdf} \approx \begin{cases} b_+ \left(\frac{\tau_{\max}}{t} \right)^{\alpha+1} & \tau_{\min} < T_w \ll t \ll \tau_{\max} \\ b_- \left(\frac{\tau_{\max}}{t} \right)^{\alpha} & \tau_{\min} \ll t \ll \tau_{\max} < T_w \end{cases} \quad (\text{S10})$$

with

$$b_{\pm} = (\alpha-1)\Gamma(\alpha) \frac{H}{\pi\gamma\sigma} \left(\frac{L_{\min}}{L_{\max}} \right)^{\alpha-1} \frac{\tau}{\tau_{\max}} \left(\frac{\alpha T_w}{\tau_{\max}} \right)^{\mu_{\pm}}, \quad (\text{S11})$$

where $\mu_+ = 1$ and $\mu_- = 0$.

Eq. (S10) shows that the distribution of the cycles length (Eq. S2) determines the intermediate asymptotic behavior of the survival probability: The power law decay changes from $t^{-\alpha-1}$ for short waiting time to $t^{-\alpha}$ when $T_w > \tau_{\max}$. Notice that this result is independent of any other parameter of the system. Thus, the behavior of the time decay of the survival probability, at saturation $T_w > \tau_{\max}$, reflects the distribution of the cycles length⁶⁸.

Table S2: simulation parameters for Fig. 3

α	1.5
γ	1.5
$L_{\max}$	2500
H	0.8
N	2^{14}

Stress	χ^2_{red}		
	Exponential	Stretched exponential	Powerlaw
1. CASP	0.008	exp	0.06
2. Gradual SHX	0.18	exp	0.45
3. LB EPEC	0.19	exp	0.21
4. LB KLY	0.63	0.15	0.05
5. Abrupt SHX	8.9	3.1	1.6
6. Sodium azide	0.31	0.053	2.7
7. CAM	0.24	0.14	0.02

Table S3. Goodness of fit tests for relaxation dynamics following various stress conditions.

Relaxation dynamics in at least 2 experiments were averaged and fitted to different functions (see **Goodness of fit tests for relaxation dynamics**). The measure of the goodness of fit is χ^2_{red} . It is smaller the closer the fitted function and the averaged lag survival function. Smallest value marked in orange. This comparison shows that relaxation dynamics from CASP, Gradual SHX, and LB EPEC stationary phase are described by an exponential function; relaxation dynamics from sodium azide is best described by stretched exponential, and Abrupt SHX, CAM, and LB KLY stationary phase are described by a power law.

“exp”: For conditions 1-3, the output for the stretched exponential was an exponential (i.e. the fit output for the stretched exponent=1).

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